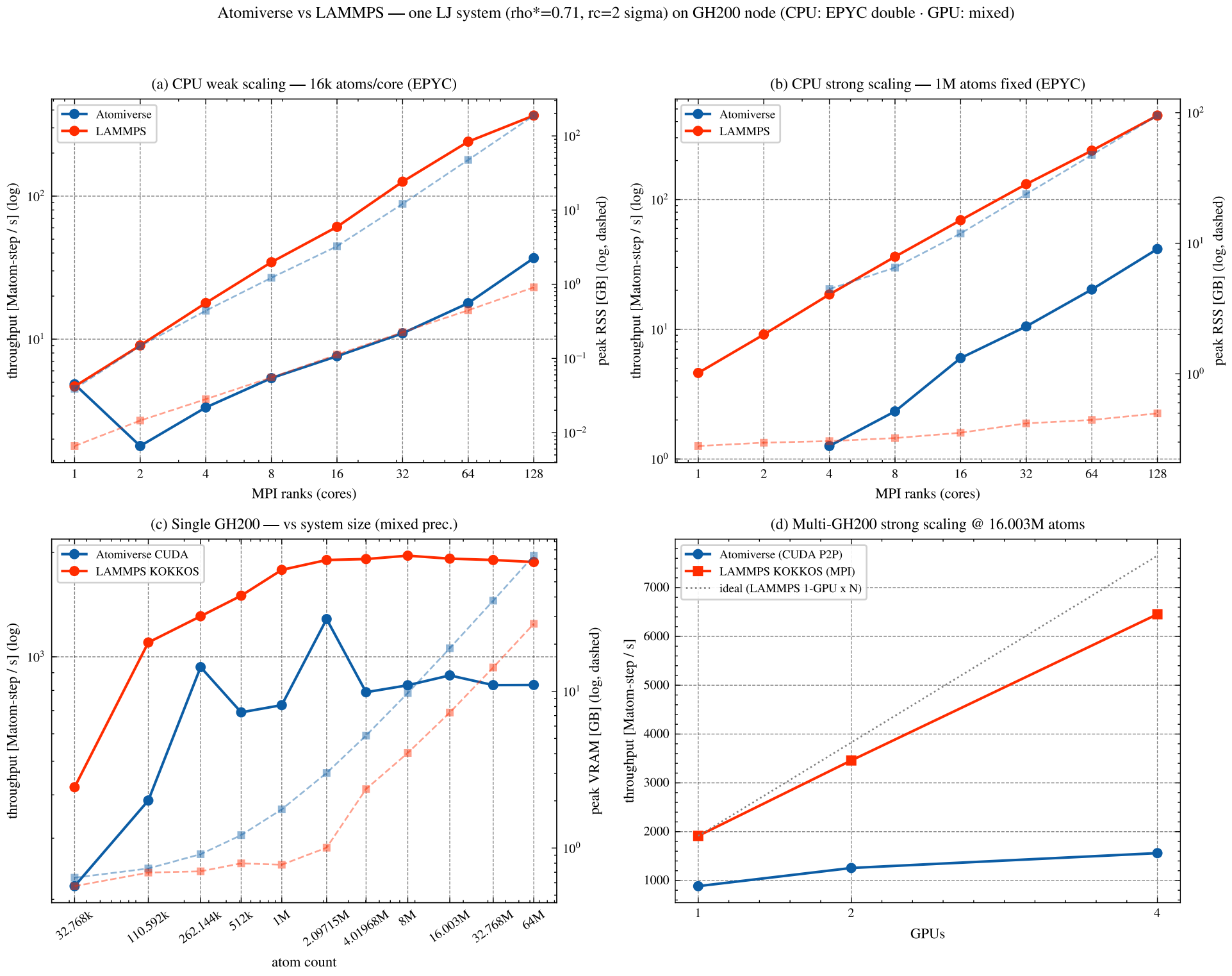
Atomiverse vs LAMMPS — one LJ system (rho*=0.71, rc=2 sigma) on GH200 node (CPU: EPYC double · GPU: mixed)
(a) CPU weak scaling — 16k atoms/core (EPYC)
Atomiverse
LAMMPS
throughput [Matom-step / s] (log)
peak RSS [GB] (log, dashed)
MPI ranks (cores)
(b) CPU strong scaling — 1M atoms fixed (EPYC)
Atomiverse
LAMMPS
throughput [Matom-step / s] (log)
peak RSS [GB] (log, dashed)
MPI ranks (cores)
(c) Single GH200 — vs system size (mixed prec.)
Atomiverse CUDA
LAMMPS KOKKOS
throughput [Matom-step / s] (log)
peak VRAM [GB] (log, dashed)
32.768k
110.592k
262.144k
512k
1M
2.09715M
4.01968M
8M
16.003M
32.768M
64M
atom count
(d) Multi-GH200 strong scaling @ 16.003M atoms
Atomiverse (CUDA P2P)
LAMMPS KOKKOS (MPI)
ideal (LAMMPS 1-GPU x N)
throughput [Matom-step / s]
GPUs